

R Variables Basis

Variables in R

Variables in *R* serve as containers for storing data values.

Creating Variables

Unlike many other languages, *R* does not require a formal variable declaration. A variable is automatically created the moment you first assign a value to it.

Use the `<-` sign to assign a value to a variable.

Example

```
# Creating variables
name <- "John"
age <- 40
```



While *R* allows `=` as an assignment operator, `<-` is strongly preferred because `=` can lead to syntax issues or unexpected behaviour in certain contexts.

Printing / Outputting Variables

R provides two primary ways to output variable values:

1. Auto-Printing

To output the variable value, just type the variable name.

Example

```
name <- "John"
```

```
name # Auto-prints variable value
```

2. Explicit Printing (`print()`)

The built-in `print()` function is available if you want to use it.

Example

```
name <- "John"

print(name) # Prints the value of the name variable
```

Explicit calling of `print()` is necessary when displaying output inside control structures (e.g., `for` loops or custom functions) where auto-printing is suppressed.

Example

```
# Example: Printing inside a loop
for (x in 1:10) {
  print(x)
}
```